

Solution

MOTION-03

Class 09 - Science

1. i. Average velocity = $\frac{\text{Total displacement travelled}}{\text{Total time taken}}$
 $= \frac{\text{Diameter of circle}}{20} = \frac{2 \times 70}{20} = 7 \text{ ms}^{-1}$
ii. Average velocity = $\frac{\text{Total distance travelled}}{\text{Total time taken}}$
 $= \frac{\text{Half circumference of circle}}{20} = \frac{\pi r}{20} = \frac{22 \times 70}{7 \times 20} = 11 \text{ ms}^{-1}$
2. Initially, the piece of cloth and the loose water in it are in a state of rest. When the cloth is shaken, it is suddenly set into motion, but the loose water in it, on account of inertia of rest continues in its state of rest. Thus water comes out of cloth in the form of fine particles.
3. The velocity time graphs have the following uses
 - i) To determine the total distance travelled by a particle.
 - ii) To determine the instantaneous velocity of the particle.
 - iii) To determine the acceleration of the object.
4. Using the equation of motion $s = ut + \frac{1}{2}at^2$
Distance travelled in 5 seconds, $s = u \times 5 + \frac{1}{2}a \times 5^2$
or $5u + \frac{25}{2}a$
Similarly, distance travelled in 4 seconds, $s' = 4u + \frac{16}{2}a$
Distance travelled in the interval between 4th and 5th seconds = $(s - s') = (u + \frac{9}{2}a)m$
5. The distance time-graph for uniform motion is a straight line not parallel to the time axis. The distance time graph for non uniform motion is not a straight line. It can be a curve or a zigzag line not parallel to time axis.
6. Given
Initial speed of train,
 $u = 90 \text{ kmh}^{-1} = 90 \times \frac{5}{18} = 25 \text{ ms}^{-1}$
Final speed, $v = 0 \text{ ms}^{-1}$
Acceleration, $a = -0.5 \text{ ms}^{-2}$
Distance covered, $S = ?$
Using the relation $v^2 - u^2 = 2aS$, we have
 $S = \frac{v^2 - u^2}{2a} = \frac{0 - (25)^2}{2 \times (-0.5)} = 625 \text{ m}$
7. Given
Initial velocity, $u = 5 \text{ ms}^{-1}$
Final velocity, $v = 0$
Acceleration in the downward direction = 10 ms^{-2}
Therefore acceleration in the upward direction $a = -10 \text{ ms}^{-2}$
Height attained by stone, $S = ?$
Time taken to attain height, $t = ?$
 - i) Using the relation : $v = u + at$
 $0 = 5 + (-10)t$ or
 $t = \frac{5}{10} = 0.5 \text{ s}$
 - ii) Using the relation : $v^2 - u^2 = 2aS$ as we have
 $S = \frac{v^2 - u^2}{2a} = \frac{0 - (5)^2}{2 \times (-10)} = 1.25 \text{ m}$
8. Uniform velocity = A body is said to have uniform velocity if it covers equal distance in equal intervals of time in a particular direction, however the time intervals may be small.
Uniform acceleration = A body is said to possess uniform acceleration if its velocity increases by equal amount in equal intervals of time, however the time intervals may be small.
9. Given
Initial velocity, $u = 0$

Acceleration, $a = 4 \text{ ms}^{-2}$

Time, $t = 10 \text{ s}$

Distance covered, $S = ?$

We know: $S = ut + \frac{1}{2} at^2$

$$S = 0 \times 10 + \frac{1}{2} \times 4 \times (10)^2$$

$$= 0 + 200 = 200 \text{ m}$$

therefore, distance covered = 200 m

10. We know that, $1 \text{ kmh}^{-1} = \frac{5}{18} \text{ ms}^{-1}$, , therefore a velocity of 120 kmh^{-1} will be $\frac{5}{18} \times 120 = 33.3 \text{ ms}^{-1}$. Now distance travelled in 30s will be $S = v \times t = 33.3 \times 30 = 999 \text{ m}$ about 1000 m.

11. Here,

Acceleration of car, $a = -6 \text{ ms}^{-2}$; Time taken to stop the car, $t = 2 \text{ s}$ and final velocity, $v = 0 \text{ ms}^{-1}$ (finally, body is at rest).

Let 'u' be the initial velocity of car.

We know that; $v = u + at$ (1)

Put the given values in equation (1), we get

$$0 = u + (-6 \text{ ms}^{-2}) \times 2 \text{ s} \text{ or } u = 12 \text{ ms}^{-1}$$

$$\text{Now, Distance covered, } S = ut + \frac{1}{2} at^2 = (12 \text{ ms}^{-1}) \times (2 \text{ s}) + \frac{1}{2} (-6 \text{ ms}^{-2}) (2 \text{ s})^2 = 24 \text{ m} - 12 \text{ m} = 12 \text{ m}$$

Thus, the car will move 12 m before it stops after the application of brakes.

12. Speed of signal = $3 \times 10^8 \text{ ms}^{-1}$

Time in which signal reaches ground = $5 \text{ min} = 5 \times 60 = 300 \text{ s}$

Distance of spaceship from the ground level = speed $\times$ time = $3 \times 10^8 \times 300 = 9 \times 10^{10} \text{ m}$.

13. i. From graph, it is clear that, velocity is not changing, therefore acceleration is equal to zero.

ii. From graph, velocity of cyclist = 20 ms^{-1}

iii. Distance covered by cyclist in 15 seconds, $s = \text{Velocity} \times \text{time} = u \times t = 20 \times 15 = 300 \text{ m}$.

14. Given total distance travelled = 200 km

$$\text{Time taken for the first half} = t_1 = \frac{S}{V_1} = \frac{100}{50} = 2 \text{ h}$$

$$\text{Time taken for the second half} = t_2 = \frac{S}{V_2} = \frac{100}{V_2}$$

$$\text{Now } v_{av} = 70 \text{ kmh}^{-1}$$

$$\text{Therefore } V_{av} = \frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{S}{t_1 + t_2} = \frac{200}{\frac{100}{50} + \frac{100}{V_2}} = 116.6 \text{ kmh}^{-1}$$

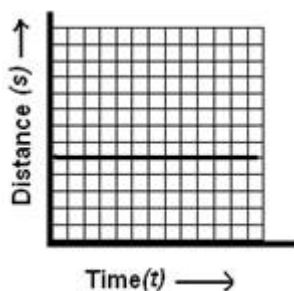
Solving for v_2 we get 116.6 kmh^{-1} .

15. i) Speed is the rate of change of motion but velocity is the rate of change of motion in a specified direction.

ii) Speed is a scalar quantity, but velocity is a vector quantity.

16. If distance travelled by an object is equal to its displacement then the magnitude of average velocity of an object will be equal to its average speed.

17. If the distance-time graph is a straight line parallel to the time axis, it indicates that with increasing time the distance travelled by that object is not increasing. Hence the object is at rest i.e. not moving.



18. Given :

Initial velocity(u) = 0,

Acceleration (a) = 2 cm/s^2 ,

time(t) = 3 s

final velocity(v)=?

Now, as we know;

$$v = u + at$$

$$v = 0 + 2 \times 3 = 6 \text{ cm/s} = 0.06 \text{ m/s}$$

Therefore the final Velocity of trolley will be 0.06m/s after start.

19. Given $v = 72 \text{ kmh}^{-1}$, $t = 10$ second, $u = 0$, $a = ?$, $v_{av} = ?$, $S = ?$

$$\text{also } v = 72 \times \frac{5}{18} = 20 \text{ ms}^{-1}$$

Using $v = u + at$ we have

$$20 = 0 + a \times 10, \text{ therefore } a = 2 \text{ ms}^{-2}$$

Using $v^2 - u^2 = 2as$ we have

$$S = \frac{v^2 - u^2}{2a} = \frac{(20)^2 - 0}{2 \times 2} = \frac{400}{4} = 100 \text{ m}$$

$$\text{Also } V_{av} = \frac{s}{t} = \frac{100}{10} = 10 \text{ ms}^{-1}$$

20. Let us assume, the final velocity with which ball will strike the ground be 'v' and time it takes to strike the ground be 't'

Initial Velocity of ball $u = 0$

Distance or height of fall $s = 20 \text{ m}$

Downward acceleration $a = 10 \text{ ms}^{-2}$

As we know,

$$2as = v^2 - u^2$$

$$v^2 = 2as + u^2$$

$$= 2 \times 10 \times 20 + 0$$

$$v^2 = 400 \text{ ms}^{-1}$$

$\therefore$ Final velocity of ball, $v = 20 \text{ ms}^{-1}$

$$t = \frac{v - u}{a}$$

$\therefore$ Time taken by the ball to strike = $\frac{20 - 0}{10}$

$$= \frac{20}{10}$$

$$= 2 \text{ seconds}$$

21. It is given that

$$u = 0 ; v = 72 \text{ km h}^{-1} = 20 \text{ m s}^{-1} \text{ and}$$

$$t = 5 \text{ minutes} = 300 \text{ s}$$

i. From Eq. we know that

$$\begin{aligned} a &= \frac{(v - u)}{t} \\ &= \frac{20 \text{ ms}^{-1} - 0 \text{ ms}^{-1}}{300 \text{ s}} \\ a &= \frac{1}{15} \text{ ms}^{-2} \end{aligned}$$

ii. From Eq. we have

$$2as = v^2 - u^2 = v^2 - 0$$

Thus,

$$\begin{aligned} s &= \frac{v^2}{2a} \\ &= \frac{(20 \text{ ms}^{-1})^2}{2 \times (1/15) \text{ ms}^{-2}} \\ &= 3000 \text{ m} \\ &= 3 \text{ km} \end{aligned}$$

The acceleration of the train is $\frac{1}{15} \text{ ms}^{-2}$ and the distance travelled is 3 km.

22. Let, The distance from point 'A' to point 'B' = $AB = x$,

Time taken to reach from A to B, when motorcyclist drives with speed $30 \text{ km h}^{-1} = t_1 = \frac{x}{30}$

Similarly, Time taken to reach from B to A, when motorcyclist drives with speed $20 \text{ km h}^{-1} = t_2 = \frac{x}{20}$

$$\text{Now, Total time} = t_1 + t_2 = \frac{5x}{60} \text{ h}$$

$$\text{Average speed for entire journey} = \frac{\text{Total distance}}{\text{Total time}} = \frac{2x}{\frac{5x}{60}} = 24 \text{ km h}^{-1}.$$

We have $0 = u - 2.5 \times 10$

Therefore $u = 25 \text{ ms}^{-1}$

27. Given $v_1 = 50 \text{ kmh}^{-1}$ and $v_2 = 490 \text{ kmh}^{-1}$

Let S be the distance. Therefore total distance travelled by the body is $= S + S = 2S$

Now time taken for the first motion $t_1 = \frac{S}{v_1} = \frac{S}{50}$

Time taken for the second motion $t_2 = \frac{S}{v_2} = \frac{S}{40}$

Total time taken $t = t_1 + t_2 = \frac{S}{50} + \frac{S}{40}$

Hence average speed $V_{av} = \frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{2S}{t} = \frac{2S}{\frac{S}{40} + \frac{S}{50}} = 44.5 \text{ kmh}^{-1}$

28. It is given that the total distance covered by Usha in 1 min is 180 m.

Displacement of Usha in 1 min = 0m

Average speed = $\frac{\text{Total distance covered}}{\text{Total time taken}}$

$$= \frac{180\text{m}}{1\text{min}} = \frac{180\text{m}}{1\text{min}} \times \frac{1\text{min}}{60\text{s}}$$

$$= 3 \text{ m s}^{-1}$$

Average velocity = $\frac{\text{Displacement}}{\text{Total time taken}}$

$$= \frac{0\text{m}}{60\text{s}}$$

$$= 0 \text{ ms}^{-1}$$

The average speed of Usha is 3 m s^{-1} and her average velocity is 0 m s^{-1}

29. Yes, if an object has moved through a distance it can have zero displacement because displacement of an object is the actual change in its position when it moves from one position to the other. So if an object travels from point A to B and then returns back to point A again, the total displacement is zero. For eg. If an object travels a distance of 6 m from point A to point B and comes back to point A again, then displacement travelled by it is $6-6 = 0\text{m}$

30. Yes, an object which has moved through a distance can have zero displacement.

Example. When a person, walking along a circular path, returns back to the starting point, after completing a circle, his displacement is zero. But he covers a distance $2\pi r$ where 'r' is the radius of circular path.

The displacement is zero, as the shortest distance between the initial and final position of the person is zero.

31. When the velocity of a body changes at a non-uniform rate, its average velocity is found by dividing the net displacement covered by the total time taken.

$$\text{i.e., Average Velocity} = \frac{\text{Net displacement}}{\text{Total time taken}}$$

In case the velocity of a body changes at a uniform rate, then the average velocity is given by the arithmetic mean of initial velocity and final velocity for a given period of time.

$$\text{i.e., Average velocity} = \frac{\text{Initial velocity} + \text{Final velocity}}{2}$$

32. Let one way distance for his trip be S .

Let t_1 be the time for his trip from home to school and t_2 be the time for his return trip.

$$\text{Then } t_1 = \frac{S}{v_1} = \frac{S}{20} \text{ h and } t_2 = \frac{S}{v_2} = \frac{S}{40} \text{ h}$$

$$\text{Therefore total time of trip is } T = t_1 + t_2 = \frac{S}{20} + \frac{S}{40} = \frac{3S}{40} \text{ h}$$

Total distance covered = $2S$

$$\text{Therefore average speed of Abdul } V_{av} = \frac{\text{total distance}}{\text{total time}} = \frac{25 \times 40}{35} = 26.7 \text{ kmh}^{-1}$$

33. Given $t = 5 \text{ s}$

Initial speed of bus

$$u = 80 \text{ kmh}^{-1} = 80 \times \frac{5}{18} = 22.2 \text{ ms}^{-1}$$

Final speed of the bus

$$v = 60 \text{ kmh}^{-1} = 60 \times \frac{5}{18} = 16.7 \text{ ms}^{-1}$$

$$\text{Now acceleration is given by the relation } a = \frac{v-u}{t} = \frac{16.7 - 22.2}{5} = -1.1 \text{ ms}^{-2}$$

34. i. A body is in uniform acceleration when equal changes in velocity take place in equal intervals of time, however small these intervals may be.

ii. A body is said to be possessing non-uniform acceleration when unequal changes in velocity take place in equal intervals of time, however small these intervals may be.

35. Velocity of an object is non-uniform if it travels unequal displacement in equal intervals of time.

$$36. (i) \text{ Average velocity} = \frac{\text{Total displacement taken}}{\text{Total time taken}}$$

$$= \frac{\text{Diameter of circle}}{20}$$

$$= \frac{2 \times 70}{20}$$

$$= 7 \text{ ms}^{-1}$$

$$(ii) \text{ Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{\text{Half circumference of circle}}{20}$$

$$= \frac{\pi r}{20}$$

$$= \frac{22 \times 70}{7 \times 20}$$

$$= 11 \text{ ms}^{-1}$$

Speed	Velocity
(i) It is the rate of change of distance.	(i) It is the rate of change of displacement.
(ii) It is a scalar quantity.	(ii) It is a vector quantity.
(iii) It is always positive.	(iii) It can be positive negative or zero.

38. Given $u = 40 \text{ ms}^{-1}$, $v = 0$, $a = -2 \text{ ms}^{-2}$, $s = ?$

Using equation $v^2 - u^2 = 2aS$ we have

$$0 = (40)^2 + 2(-2)S \text{ or } 4S = 1600 \text{ or } S = 400 \text{ m}$$

Thus the train stops in 400 m. Since the platform is 400 m long therefore the train just stops in time.

39. Initial speed of bus (u) = $80 \text{ km h}^{-1} = \frac{80 \times 100}{60 \times 60} \text{ seconds} = \frac{200}{9} \text{ ms}^{-1} = 22.22 \text{ ms}^{-1}$

final speed of bus (v) = $60 \text{ km h}^{-1} = \frac{60 \times 100}{60 \times 60} \text{ seconds} = \frac{50}{3} \text{ ms}^{-1} = 16.67 \text{ ms}^{-1}$ time (t) = 5 s

$$\text{acceleration (a)} = \frac{(v-u)}{t} = \frac{(16.67-22.22)}{5} = \frac{-5.55}{5} = -1.11 \text{ ms}^{-2}$$

40. Given $t = 10 \text{ min} = 10 \times 60 = 600 \text{ s}$

Initial speed of train, $u = 0 \text{ ms}^{-1}$

$$\text{Final speed of train } v = 40 \text{ kmh}^{-1} = 40 \times \frac{5}{18} = 11.1 \text{ ms}^{-1}$$

$$\text{Now acceleration is given by the relation } a = \frac{v-u}{t} = \frac{11.1-0}{600} = 0.0185 \text{ ms}^{-2}$$

41. Given $r = 42,250 \text{ km}$, $T = 24 \text{ hours} = 86400 \text{ s}$

The linear velocity is given by the expression

$$V = \frac{2\pi r}{t} = \frac{2 \times 3.14 \times 42250}{86400} = 3.07 = 3.1 \text{ kms}^{-1}$$

42. Initial odometer reading = 2000 km

Final odometer reading = 2400 km

Distance covered by the car

$$s = 2400 \text{ km} - 2000 \text{ km} = 400 \text{ km}$$

Time elapsed, $t = 8 \text{ h}$

$$\text{The average speed of the car is, } v = \frac{s}{t} = \frac{400 \text{ km}}{8 \text{ h}}$$

$$= 50 \text{ km h}^{-1}$$

$$= 50 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}}$$

$$= 13.9 \text{ ms}^{-1}$$

The average speed of the car is 50 km h^{-1} or 13.9 m s^{-1}

43. Here, Initial velocity of car, $u = 0$

$$\text{By Using the relation, } S = ut + \frac{1}{2}at^2$$

$$\text{The distance travelled in first 8s, } S_1 = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 5 \times 8^2 = 160 \text{ m}$$

$$\text{At this point the velocity, } v = u + at = 0 + 5 \times 8 = 40 \text{ ms}^{-1}.$$

Since the velocity remains constant, So, the distance covered in last four seconds, $S_2 = \text{constant velocity} \times \text{time} = 40 \times 4 = 160 \text{ m}$

$$\text{Hence, total distance, } S = S_1 + S_2 = 160 \text{ m} + 160 \text{ m} = 320 \text{ m}$$

44. If the velocity is uniform then,

Acceleration $a = 0$

So, the equations of motion will become:

First equation: $v = u$

Second equation: $s = ut$

Third equation: $v^2 - u^2 = 0$

45. Given

Diameter of circular track, $2r = 200$ m

Circumference of circular track = $2\pi r$

$$S = 2\pi r = \frac{22}{7} \times 200 = \frac{4400}{7} \text{ m}$$

Time for completing one round = 40 s.

Time for which the athlete ran = 2 min and 20 s = 140 s.

Now distance covered by the athlete in 40 s

$$S = \frac{4400}{7} \text{ m}$$

i) Therefore, distance covered by athlete in 140 s = $\frac{4400}{7} \times \frac{140}{40} = 2200$ m

ii) As the athlete returns to the initial point in 40 s, this displacement = 0

Now

Number of rounds in 40 second = 1

Hence number of rounds in 140 s is = $\frac{140}{40} = 3.5$

For each complete round the displacement is zero.

Therefore for 3 complete rounds, the displacement will be zero.

The final displacement will be due to half the round.

In half round distance covered = half of circumference.

Thus, his displacement = diameter of circular track = 200 m

Displacement after 140 s = 200 m

46. Uniform velocity: An object with uniform velocity covers equal distance in equal intervals of time in a specified direction. For e.g. an object moving with speed of 40 km/hr towards west has uniform velocity.

Non-uniform velocity: When an object covers unequal distances in equal intervals of time in a specified direction, or if the direction of motion changes, it is said to be moving with a non-uniform or variable velocity. For e.g. revolving fan at a constant speed has variable velocity as direction of blades of fan is continuously changing.

47. Given

Initial speed of bus, $u = 0 \text{ ms}^{-1}$

Final speed of bus, $v = ?$

Acceleration, $a = 0.1 \text{ ms}^{-2}$

Time, $t = 2 \text{ min} = 120 \text{ s}$

Distance travelled, $s = ?$

i) We know, $v = u + at$

$$\text{or } v = 0 + 0.1 \times 120 = 12 \text{ ms}^{-1}$$

ii) To find distance travelled, we use the equation

$$S = ut + \frac{1}{2} at^2$$

$$\text{or } S = 0 \times 120 + \frac{1}{2} \times 0.1 \times (120)^2 = 720 \text{ m}$$

Therefore

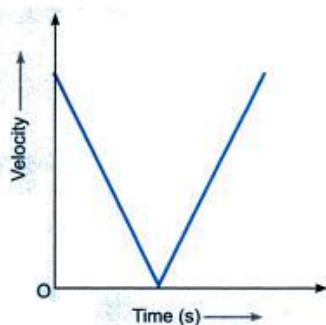
Final speed acquired = 12 ms^{-1}

Distance travelled = 720 m

48. Speed = Slope of distance-time graph.

The smaller the slope, the smaller is the speed. From the figure, slope is minimum for car D. So, D is the slowest car.

49.



50. Since, the object is dropped from rest, Its initial velocity = '0' m/s.

Initial difference in height = (150 - 100) m = 50 m

By using , equation, $S = ut + \frac{1}{2} at^2$, we have,

Distance travelled by first body in 2 s = $h = 0 + \frac{1}{2} g(2)^2 = 2g$ (For downward motion, g = Positive)

Distance travelled by another body in 2 s = $h_2 = 0 + \frac{1}{2} g(2)^2 = 2g$

After 2s, height at which the first body will be = $h'_1 = 150 - 2g$

After 2 s, height at which the second body will be = $h'_2 = 100 - 2g$

Thus, after 2 s, difference in height = $150 - 2g - (100 - 2g) = 150 - 2g - 100 + 2g = 50 \text{ m} = \text{initial difference in height.}$

Therefore, difference in height does not vary with time.

51.

Distance	Displacement
(i) It is the length of the actual path traversed between its initial and final positions.	(i) Displacement is the shortest path between the initial and the final positions of the particle.
(ii) It is a scalar quantity.	(ii) It is a vector quantity.
(iii) It can never be negative.	(iii) It can be positive, negative or zero.

52. As the motion is uniform therefore the velocity of the body will remain same even after 10s. Therefore the velocity of the body will be 10 ms^{-1} .

53. Let at time $t = 0$, body has initial velocity = V_0

At time 't', body has final velocity = V

s = distance traveled in time 't'

We know, total distance travelled = Average velocity x time

$$\text{Average velocity} = \frac{\text{initial velocity} + \text{final velocity}}{2}$$

$$= \frac{V_0 + V}{2}$$

$$\text{Total distance} = s = \frac{V_0 + V}{2} \times t$$

$$2s = (V_0 + V) t \dots\dots\dots (i)$$

Now from first equation of motion,

$$V = V_0 + at \dots\dots\dots (ii)$$

On putting the value of V in eqⁿ (i), we get

$$2s = (V_0 + V_0 + at) t$$

$$2s = 2V_0 t + at^2$$

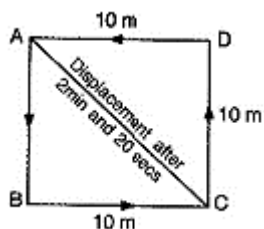
$$s = V_0 t + \frac{1}{2} at^2$$

Let $V_0 = u$

$$s = ut + \frac{1}{2} at^2$$

54. The perimeter of square field ABCD = $4 \times 10 \text{ m} = 40 \text{ m}$

Time for moving around the 10 m square field once = 40 s.



Time for the journey of farmer = 2 min and 20 s = 140 s Number of times the farmer moves around the square field = $\frac{140}{40} = 3.5$ times

For going once and thrice around the square field, the displacement = 0

For going $\frac{1}{2}$ times the square field, the distance covered = $40\text{m} \times \frac{1}{2} = 20\text{m}$

It is obvious from the figure, that if the farmer starts from pt. A then he will cover 10 m along AB and then 10 m along BC.

Therefore displacement of a farmer from the pt. A to point C is

$$AC = \sqrt{BC^2 + AB^2} = \sqrt{10^2 + 10^2} = 14.14\text{m}$$

55. We know $1 \text{ kmh}^{-1} = \frac{5}{18} \text{ ms}^{-1}$

Initial speed, $u = 80 \text{ kmh}^{-1} = 80 \times \frac{5}{18} \text{ ms}^{-1}$

$$= \frac{200}{9} \text{ ms}^{-1}$$

Final speed, $v = 60 \text{ kmh}^{-1} = 60 \times \frac{5}{18} \text{ ms}^{-1}$

$$= \frac{50}{3} \text{ ms}^{-1}$$

Time, $t = 5\text{s}$

Acceleration of bus, $a = \frac{v-u}{t}$

$$= \frac{\frac{50}{3} \text{ ms}^{-1} - \frac{200}{9} \text{ ms}^{-1}}{5\text{s}}$$

$$= \frac{-50 \text{ ms}^{-1}}{9 \times 5\text{s}}$$

$$= -\frac{10}{9} \text{ ms}^{-2}$$

[-ve acceleration means it is retardation i.e., speed of the body is decreasing]

56. Initial velocity of body, $u = 0.5 \text{ m/s}$

Final velocity of body, $v = 0$

Negative acceleration, $a = -0.05 \text{ m/s}^2$

Now, from the first equation of the motion, we have

$$v = u + at, \text{ Where } t = \text{time}$$

Put the given values, we get

$$0 = 0.5 + (-0.05t) \Rightarrow 0.5 = 0.050t$$

$$\therefore t = 0.5/0.05 = 10 \text{ s}$$

Thus, the body will take 10 s to stop.

57. We need to calculate the time taken in each of the trip. After that, we can calculate the average speed.

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

$$\therefore = \frac{s}{t_1}$$

$$\Rightarrow 20 \text{ km/h} = \frac{s}{t_1}$$

$$\Rightarrow t_1 = \frac{s}{20} \text{ h}$$

$$\Rightarrow 30 \text{ km/h} = \frac{s}{t_2}$$

$$\Rightarrow t_2 = \frac{s}{30} \text{ h}$$

$$(t_1 + t_2) = \frac{s}{20} + \frac{s}{30}$$

$$\Rightarrow (t_1 + t_2) = \frac{5s}{60} = \frac{s}{12} \text{ h}$$

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

$$= \frac{2s}{\frac{s}{12}}$$

$$= 2 \times 12$$

$$= 24 \text{ km/hr}$$

58. Distance covered by the satellite in 24 hours.

$$S = 2\pi r = 2 \times \frac{22}{7} \times 42250$$

$$= 265571.43 \text{ km}$$

Therefore speed of satellite

$$v = \frac{\text{distance travelled}}{\text{time taken}} = \frac{265571.43}{24 \times 60 \times 60} = 3.07 \text{ kms}^{-1}$$

59. Velocity = $\frac{\text{Distance}}{\text{time}}$ or distance = velocity $\times$ time

$$\text{Therefore } S = \frac{120 \times 1000}{3600} \times 30 = 1000 \text{ m}$$

60. Initial velocity of boat, $u = 0$

$$\text{Acceleration, } a = 3.0 \text{ ms}^{-2}$$

$$\text{Time, } t = 8 \text{ s}$$

$$\text{Distance covered, } S = ?$$

$$\text{Using the relation } S = ut + \frac{1}{2} at^2 \text{ we have}$$

$$S = 0 \times 8 + \frac{1}{2} \times 3 \times 8^2 = 96 \text{ m}$$

61. BASIS FOR COMPARISON	DISTANCE	DISPLACEMENT
Meaning	Distance refers to the amount of space between two points, measured along the actual path, linking them.	Displacement refers to the amount of space between the two points, measured along the minimum path linking them.
What is it?	Length of the total avenue traversed by the body.	Least distance between starting and ending point.
Quantity	Scalar Quantity	Vector Quantity
Information	Gives complete information of the route followed by the body.	Does not give complete information of the route followed by the body.
Time	Distance can never decrease with time.	Displacement can decrease with time.
Values	Positive	Positive, Negative or Zero
Unique path	No	Yes
Denoted by	d	s
Formula	Speed $\times$ Time	Velocity $\times$ Time

62. Side of square = 10 m

$$\text{Perimeter of square} = 4 \times \text{Side} = 4 \times 10 \text{ m} = 40 \text{ m}$$

Let farmer start moving from A which is treated as a reference point. He covers 40 m in 40 s. Therefore, distance travelled by farmer in 2 minutes and 20 seconds i.e., 140 seconds = 140 m. The farmer, therefore completes 3 rounds of field and 20 metres more distance.

Thus, position of farmer at the end of 140 seconds will be C.

Therefore, the magnitude of displacement of the farmer at the end of 140 s = Shortest distance between initial and final position =

$$AC = \sqrt{AB^2 + BC^2} \dots\dots\dots (\text{pythagorus theorem})$$

$$= \sqrt{10^2 + 10^2}$$

$$= \sqrt{200}$$

$$= 10\sqrt{2} \text{ m} = 10 \times 1.414 = 14.14 \text{ m}$$

$$63. \text{ Velocity} = \frac{18}{1000} \times \frac{1}{\frac{1}{3600}} = 64.8 \text{ kmh}^{-1}$$

64. Given

$$\text{Initial velocity, } u = 0$$

$$\text{Final velocity, } v = ?$$

$$\text{Time, } t = 3 \text{ s}$$

$$\text{Acceleration, } a = 2 \text{ cms}^{-2}$$

We know that $v = u + at$

$$\text{or } v = 0 + 2 \times 3 = 6 \text{ cms}^{-1}$$

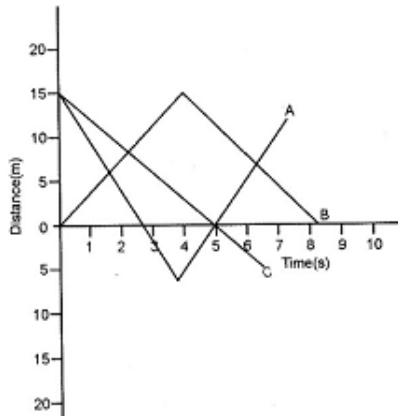
65. Given $u = 5 \text{ ms}^{-1}$, $g = -10 \text{ ms}^{-2}$, $h = ?$ and $t = ?$ at the highest point the velocity of the stone will be zero, therefore $v = 0 \text{ ms}^{-1}$

Using the equation $v^2 - u^2 = 2aS$ we have $0^2 - 5^2 = 2 \times (-10) \times h$,

$$\text{therefore } h = \frac{20}{20} = 1.25 \text{ m}$$

$$\text{Using the equation } v = u + gt \text{ we have } t = \frac{v - u}{g} = \frac{0 - 5}{-10} = 0.5 \text{ s}$$

66. When the displacement is zero, it does not mean that distance is also zero. Displacement can be zero when the moving object comes back to its original position. Displacement is either equal to or less than distance but the distance traveled is always more than zero.
67. i. GRAPH A: The displacement of the body is 12 m and the distance travelled is 37 m
 ii. GRAPH B: Displacement is zero and the distance travelled is $15 + 15 = 30 \text{ m}$
 iii. GRAPH C: Displacement is about 5 m and distance travelled is about 20 m.



Graph C represents a motion with negative acceleration.

68. Speed = Slope of distance - time graph. The smaller the slope, the smaller is the speed.

69. Height = h

$$\text{Distance} = s = 320 \text{ m}$$

$$\text{Acceleration due to gravity} = g = 10 \text{ m/s}^2$$

$$\text{Initial velocity} = u = 0$$

$$(i) \text{ from } s = ut + \frac{1}{2}at^2$$

$$h = ut + \frac{1}{2}gt^2$$

$$320 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\frac{320 \times 2}{10} = t^2$$

$$64 = t^2 \quad t = \sqrt{64}$$

$$t = 8 \text{ sec}$$

$$(ii) \text{ from } v = u + at$$

$$v = 0 + 10 \times 8$$

$$v = 80 \text{ m/s}$$

70. a. $u = 0$, $a = 0.1 \text{ ms}^{-2}$, $t = 2 \text{ min} = 120 \text{ seconds}$.

$$v = u + at = 0 + 0.1 \times 120 = 12 \text{ ms}^{-1}$$

$$\text{so speed acquired} = v = 12 \text{ ms}^{-1}$$

$$b. S = 0 \times 120 + \frac{1}{2} \times 0.1 \times 120^2 = 720 \text{ m.}$$

71. Given, initial velocity of electron, $u = 5 \times 10^4 \text{ ms}^{-1}$ and acceleration, $a = 10^4 \text{ ms}^{-2}$

$$i. \text{ Final velocity of electron} = v = 2 \times u = 2 \times 5 \times 10^4 \text{ ms}^{-1}$$

$$\text{We know that, } v = u + at \text{ or } t = \frac{v - u}{a} = \left(\frac{10 \times 10^4 - 5 \times 10^4}{10^4} \right) = \frac{5 \times 10^4}{10^4} = 5 \text{ s}$$

$$ii. \text{ Using relation; } S = ut + \frac{1}{2}at^2$$

$$\text{Distance covered by electron in the given time, } S = (5 \times 10^4) \times 5 + \frac{1}{2}(10^4) \times (5)^2 = (25 \times 10^4 + \frac{25}{2} \times 10^4) = 37.5 \times 10^4 \text{ m}$$

72. Speed of the cheetah in ms^{-1} is $90 \times \frac{5}{18} \text{ ms}^{-1} = 25 \text{ ms}^{-1}$

$$\text{Distance of the prey} = 100 \text{ m}$$

$$\text{Therefore time to catch the prey} = \frac{\text{Distance}}{\text{Speed}} = \frac{100}{25} = 4S$$

73. Here, initial velocity, $u = 90 \text{ kmh}^{-1}$

$$\begin{aligned} &= \frac{90 \text{ km}}{1 \text{ h}} \\ &= \frac{90 \times 1000 \text{ m}}{60 \times 60 \text{ s}} \\ &= 25 \text{ ms}^{-1} \end{aligned}$$

Acceleration, $a = -0.5 \text{ ms}^{-2}$

Final velocity $v = 0$

From the equation of motion, $2as = v^2 - u^2$

$$\begin{aligned} s &= \frac{v^2 - u^2}{2a} \\ &= \frac{0 - (25 \text{ ms}^{-1})^2}{2 \times (-0.5 \text{ ms}^{-2})} \\ &= \frac{-625 \text{ m}^2 \text{ s}^{-2}}{-1.0 \text{ ms}^{-2}} \\ &= 625 \text{ m} \end{aligned}$$

The train will go 625 m further, after applying the brakes.

74. a. Distance covered $= \pi \times OA = \pi \times 5 = 5\pi \text{ cm}$

b. Displacement $= 2 \times OB$
 $= 2 \times 5 = 10 \text{ cm}$ along AB

75. The shape of the path followed by the ball is parabolic.

Vertically downward motion is due to gravitational acceleration. Horizontal motion is due to zero acceleration.